# BRAYDEN HUGUENARD

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## EDUCATION

**GEORGIA INSTITUTE OF TECHNOLOGY | GPA: 4.0/4.0** *Atlanta, Georgia*

**M.S. Computer Science | Machine Learning** *Graduation: May 2028*

**B.S. Computer Science | Intelligence & Modeling and Simulation** *Graduation: December 2026*

- Relevant Courses: ML, Intro to AI, Computer Vision, Deep Learning, Databases, Data Structures & Algorithms, Linear Algebra
- Recipient of Zell Miller (2024, 2025, 2026), Daniel Hager (2024), and Delta Air Lines Academic (2025, 2026) Scholarships

## SKILLS

**Programming Languages:** Python, Swift, SQL, Java, C, C++, JavaScript, HTML

**Frameworks & Libraries:** SwiftUI, React, FastAPI, PyTorch, scikit-learn, LangChain, FAISS, Pandas, NumPy, SwiftData, CoreML

**Cloud, Tools & Platforms:** AWS (Lambda, Bedrock, S3, ECS, CloudWatch), MongoDB, Terraform, Docker, Firebase, Supabase

## EXPERIENCE

**Technical Lead | *Georgia Tech App*** *December 2025 – Present*

- Architected SwiftUI codebase and established custom SwiftLint standards, aligning a 25+ person team on a unified approach
- Integrated 6 GT services across dining, transit, capacity, and events through 13+ API endpoints and ingestion pipelines
- Conducted user interviews to uncover usability issues and translate user feedback into product design and feature decisions
- Built adaptive polling for live bus tracking, reducing unnecessary network requests while maintaining real-time ETA updates
- Integrated GT SSO (CAS/OIDC) for secure backend access and built API pipelines for live dining, capacity, and event data
- Led collaboration across SGA, OIT, Dining, Library, and Institute Communications to integrate campus services into the app
- Partnered with SGA as the app helped shape a new committee supporting student-built technology projects across campus

**Teaching Assistant | *CS4400 – Intro to Database Systems*** *August 2026 – Present*

- Develop and administer database coding exams for 500+ students and host weekly office hours and technical review sessions
- Evaluate semester-long database projects through required demos, providing feedback on design and implementation

**AI Innovation Intern | *Assent*** *May 2026 – August 2026*

- Independently owned an internal LLM evaluation platform supporting 15 regulations, from prototype through deployment
- Built offline evaluation tool benchmarking model changes across 90+ labeled documents using accuracy and F1
- Improved evaluation accuracy from 64% to 81% through iterative criteria and methodology refinement with SMEs
- Reduced per-document token usage ~40%, from ~19K to ~11K, by consolidating common criteria and skipping redundant checks
- Evaluated Lambda and FastAPI architectures, deploying FastAPI on ECS with Terraform to support differing team requirements
- Built a 248-test hermetic suite covering end-to-end evaluation, API, and data-processing workflows with a 100% pass rate
- Connected three engineering teams with compliance SMEs, driving collaboration and adoption of the evaluation platform

**Software Engineer | *PlayOn Sports (Client Capstone)*** *January 2026 – April 2026*

- Developed a Tree-of-Thought text-to-SQL pipeline generating 3 parallel queries and using an LLM judge to select the best
- Built SQL safety validation blocking 9 write/DDL operations before execution, enforcing read-only queries with a 200-row cap
- Engineered dynamic schema inference and chunked concurrent ingestion for CSV datasets exceeding 600K rows
- Built a custom SVG chart-rendering engine with axis scaling and chart geometry, enabling one-click PDF report generation
- Placed 2nd overall among 40+ teams at the GT Capstone Expo for technical execution and project presentation
- Improved text-to-SQL accuracy by 23 points, from 68% to 91%, through targeted prompt engineering for complex query handling

**Teaching Assistant | *CS3600 – Intro to Artificial Intelligence*** *August 2025 – May 2026*

- Led 3 TAs developing review materials for 600+ students across two semesters, including practice resources and course reviews
- Held weekly office hours teaching AI concepts while reviewing and improving homework assignments, exams, and solutions

**Technical Lead | *Georgia Tech iOS Club*** *July 2025 – November 2025*

- Led and mentored 25+ developers building Reverie, winning “Best App” and “Best Design” at Demo Day
- Built an on-device AI pipeline using Apple Foundation Models and Stable Diffusion/CoreML to generate dream imagery
- Accelerated on-device image generation ~6×, from 30+ seconds to ~5 seconds, through model caching and concurrency
- Built a Vision-based image pipeline to create stickers from generated dreams, with persistent image storage through Firebase

**Developer | *Georgia Tech iOS Club*** *August 2024 – December 2024*

- Built Nomad’s stop-finding experience in SwiftUI, integrating Yelp’s API with typed Codable models and search filters
- Built itinerary workflows for adding, removing, and geocoding stops, contributing 55+ commits to a 20+ developer codebase
- Won “Best App” and “Most Popular” at iOS Club Demo Day for Nomad, a collaborative road-trip planning application

## PROJECTS

**Teen Mental Health Predictive Model** *September 2025 – December 2025*

- Diagnosed a synthetic dataset with no learnable signal and rebuilt the ML pipeline on validated teen depression data
- Built and tuned Random Forest and SVM models achieving ~85% accuracy and ~0.88 F1 with cross-validation
- Identified 2 dominant Random Forest features that preserved model performance while reducing training time by ~80%

**MLB Season Simulator** *June 2024 – August 2024*

- Built a probabilistic MLB engine using real player statistics to simulate at-bats, baserunner advancement, and game outcomes
- Integrated 30+ real-world datasets to simulate 2,430-game seasons, standings, playoff seeding, and postseason play